

Abandoning Hunting and Gathering

For much of human history, people survived by hunting and gathering, but in the Neolithic period, starting around 10,000 B.C.E., this was gradually given up in favor of growing grains and other plants or domesticating animals (pastoralism). Why did early societies in so many parts of the world gradually abandon a way of life based on food gathering? Some theories assume that people were drawn to food production by its obvious advantages. For example, it has recently been suggested that people settled in what is now the Middle East so they could grow enough grains to ensure themselves a ready supply of beer. Beer drinking is frequently depicted in ancient Middle Eastern art and can be dated to as early as 3500 B.C.E.

However, most researchers today believe that climate change drove people to abandon hunting and gathering in favor of pastoralism and agriculture. So great was the global warming that ended the great Ice Age that geologists give the next era which began around 9000 B.C.E. a new name: the Holocene. Scientists have also found evidence that temperate lands were exceptionally warm between 6000 and 2000 B.C.E., the era when people in so many parts of the world adopted agriculture. The precise nature of the crisis probably varied. In the Middle East taking up food production may have been a response to shortages of wild food caused by a dry spell or population growth. Elsewhere, a warmer, wetter climate could have promoted rapid forest growth in former grasslands, reducing the supplies of game and wild grains.

Additional support for an ecological explanation comes from the fact that in many drier parts of the world, where wild food remained abundant, agriculture was not adopted. The inhabitants of Australia continued to rely exclusively on foraging until recent centuries, as did some peoples in all the other continents. Many Amerindians in the arid grasslands from Alaska to the Gulf of Mexico hunted bison, while in the Pacific Northwest others took up salmon fishing. Abundant supplies of fish, shellfish, and aquatic animals permitted food gatherers east of the Mississippi River in North America to become increasingly sedentary. In the equatorial rain forest and in the southern part of Africa, conditions favored retention of the older ways. The reindeer-based societies of northern Eurasia were also unaffected by the spread of farming.

Whatever the causes, the effects of the gradual adoption of food production in most parts of the world were momentous. A hundred thousand years ago, there were fewer than 2 million people, and their range was largely confined to the temperate and tropical regions of Africa and Eurasia. During the last glacial epoch, between 32,000 and 13,000 years ago, human populations may have fallen even lower. Then, as the glaciers retreated, people moved into new land and adopted agriculture. Their numbers gradually rose to 10 million by 5000 B.C.E. and mushroomed to between 50 million and 100 million by 1000 B.C.E. This increase brought important changes to social and cultural life.

The evidence that an ecological crisis may have driven people to food production has led researchers to reexamine the assumption that people in agricultural societies were better off than foragers. Modern studies demonstrate that food producers have to work much harder and for much longer periods than do food gatherers. In return for modest harvests, early farmers needed to put in long days of arduous labor clearing and cultivating the land. Pastoralists had to guard their herds from wild predators, guide them to fresh pastures, and tend to their many needs. There is also evidence that even though the food supply of early farmers was more secure than that of food-gathering peoples and pastoralists, the farmers' diet was less varied and nutritious. Skeletal remains show that on average Neolithic farmers were shorter than earlier food-gathering peoples and more likely to die at an earlier age from contagious diseases. People in permanent settlements were more exposed to diseases from water contaminated by human waste, to disease-bearing vermin and insects that infested their bodies and homes, and to new diseases that migrated from their domesticated animals (especially pigs and cattle).